

Understanding Tool–Tissue Interactions in Surgical Videos

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Abstract—Artificial intelligence (AI) is increasingly adopted in surgery for performance assessment and intraoperative decision support. A key capability in this context is the automatic recognition of tool–tissue interactions (TTIs) from surgical videos. Existing approaches for modelling surgical actions, mainly represented by $\langle \text{tool, verb, tissue} \rangle$ triplets, do not explicitly model contact. In this work, we propose a formulation of the TTI understanding problem based on contact-aware interaction modelling, which introduces two sequential objectives: (i) spacial detection of physical contact between tool and tissue, and (ii) classification of the interaction type. By explicitly incorporating depth information, our approach moves beyond purely 2D semantic reasoning and enables inference of actual physical contact rather than visual co-occurrence. This formulation enables the use in robotic and safety-critical applications for which semantic understanding only is not sufficient. The proposed architecture integrates YOLO-based instance segmentation, depth estimation, and a Vision Transformer for interaction reasoning. Experimental results show that the proposed method achieves 87.61% end-to-end accuracy on the TTI understanding task, with F1 score of 88.03%, demonstrating the feasibility of reliable, contact-aware TTI inference.

I. INTRODUCTION

Surgical complications often arise from misidentification of anatomy leading to intraoperative errors. Artificial intelligence (AI) has shown promise in supporting surgery through endoscopic video analysis for surgical quality assessment and improvement. Laparoscopic cholecystectomy (LC), one of the most common abdominal surgical procedures worldwide, exemplifies the need for such AI assistance: while LC is generally safe, iatrogenic injuries, particularly bile duct injuries, still occur due to errors in recognizing anatomical structures or misjudging dissection planes [1], [2]. To this end, many existing works in the literature leverage AI and computer vision methods for laparoscopic procedures assistance. They aim at providing objective performance assessment [3], offering real-time intraoperative assistance [4], [5], [6], [7], [8] and supporting surgical training. Among these, Murali et al. [9] used Graph Neural Networks to assess the Critical View of Safety and prevent bile duct injuries; Golany et al. [10] employed multi-stage temporal

convolutional networks to recognize surgical phases; Endo et al. [11] proposed YOLOv3-based anatomical structure detection; and Madani et al. [12], Laplante et al. [13], and Protserov et al. [14] developed GoNoGoNet to identify safe and unsafe zones during laparoscopic cholecystectomy procedures. On the other hand, Nwoye et al. [15] have further advanced instance-based tool detection and tracking. While these approaches detect tools or anatomy, they do not infer actual interaction, leaving the understanding of whether a tool acts on a region to human judgment.

A critical challenge is, then, to derive clinically-meaningful interpretation of surgeon behaviour and actions, such as accurately recognizing Tool–Tissue Interactions (TTIs), especially when monocular cameras are used. Beyond safety, such recognition enables objective performance assessment, providing metrics and detailed analytics for feedback, credentialing and coaching. Developing AI systems capable of accurate TTI recognition can therefore reduce risks during surgery and improve the systematic evaluation of surgical performance.

In the literature, the general surgical activity recognition problem has been widely studied under the formulation of surgical action triplets, which represent the dominant paradigm for modelling activities in surgical videos. Surgical action triplets are structured representations of fine-grained surgical activities, typically expressed as $\langle \text{instrument, verb, tissue} \rangle$, answering the question: “Which tool performs which action on which anatomical structure?” [16].

Early works formalized triplet recognition as a frame-level classification task. TripNet predicts instruments, verbs, and targets jointly using multi-task learning, introducing an instrument-centric Class Activation Guide and a 3D interaction space to model valid triplet associations [16]. Building on this, the challenges CholecTriplet2021 and CholecTriplet2022 [17], [18] standardized triplet recognition as a community benchmark, evaluating various architectures, including attention-based and temporal models, and highlighting the difficulty of multi-component association and the importance of estimating also the spatial locations of the triplets. Subsequent works moved toward spatially grounded triplets. For instance, [19] links triplets to specific instrument instance masks, enabling pixel-level interpretability.

Across these efforts, the trend moves from frame-level recognition to spatially localized and instance-level grounded triplets. Explicit physical contact between the tool and tissue is, however, not modelled nor required. As a result, triplets may predict interactions even when the tool and tissue are spatially proximate but not actually in contact: an assumption

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that is often acceptable for semantic understanding but limiting for downstream robotic or safety-critical applications.

The objective of the present work is fundamentally different: it explicitly focuses on contact-aware interaction modelling introducing a new *TTI understanding* task which involves *detecting* the occurrence of TTI events, with segmentation of tool and tissue involved in the event, *identifying* the type of interaction, and *isolating* the tool-tissue pairs that are actually in contact.

Relying on a dataset of approximately 14,000 annotated frames that link instrument and tissue interaction instance masks with tool and interaction types, we incorporate depth information into the learning process, enabling reasoning about actual physical contact rather than mere visual overlap or co-occurrence. This choice is central to the proposed contribution and cannot be achieved within the standard triplet formulation without additional sensing or modelling assumptions. The representation here proposed super-classes tools and tissues by abstracting away fine-grained tool and tissue categories, which are the focus of triplet recognition, and instead emphasizes whether contact truly exists, and what type of physical interaction is occurring.

To summarize, the proposed contribution represents the missing link between explored tasks, like tool and anatomy detection/segmentation, and the explicit understanding of surgical interactions, bridging a critical gap in surgical AI. In the context of surgical robotics this capability is essential for implementing context-aware safety mechanisms, such as active constraints, which can inhibit robot motion or trigger alerts when an unintended physical contact occurs. Furthermore, the precise classification of interaction types (e.g., distinguishing retraction from dissection) enables closed-loop automation, allowing the robotic system to verify the successful execution of a surgical sub-task (e.g., confirming tissue grasping) before transitioning to the next action. Being compatible with standard monocular cameras, the proposed pipeline can enhance situational awareness in current robotic platforms without requiring additional sensors.

The remainder of the paper is organized as follows: Section II describes the proposed AI pipeline for the TTI understanding task; Section III details the dataset used for training the AI models; Section IV discusses the obtained results; finally, Section V draws conclusions of the work.

II. METHODOLOGY

In this work we propose a new *TTI understanding* task which involves *detecting* the occurrence of TTI events, with segmentation of tool and tissue involved in the event, *identifying* the tool and the interaction type, *isolating* the tool-tissue pairs that are actually in contact.

Surgical videos are evaluated frame-by-frame using the pipeline shown in Fig. 1. This kind of architecture enables the TTI understanding process to run in real-time, thus providing both post-hoc analysis and intra-operative assessments. The corresponding code is available on Github¹.

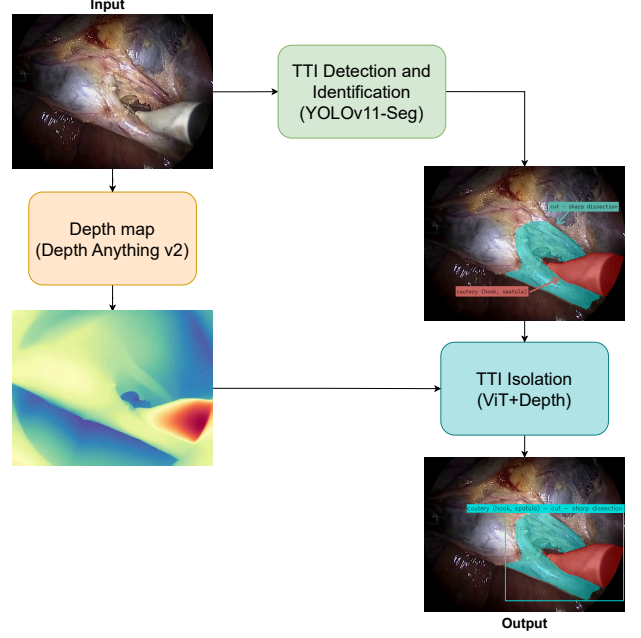


Fig. 1: Pipeline of the TTI understanding process.

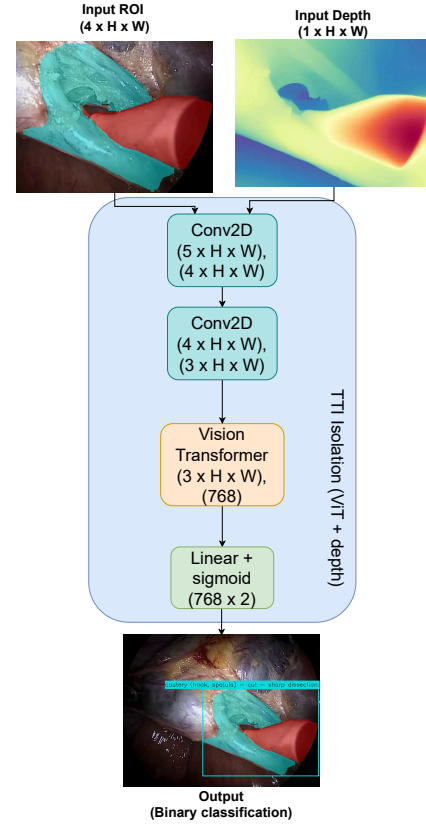


Fig. 2: Architecture of the TTI Isolation network.

¹Repository link: <https://github.com/Carlo1109/TTI.Detection>

The proposed pipeline, processes each frame by first passing it through YOLOv11-Seg model [20], fine-tuned using the dataset described in Sec. III. This model outputs the segmentation masks for the detected tools and the tissue regions that are interacting with some tool, together with their respective predicted class (i.e., the type of tool and the type of interaction in which the tissue portion is involved), enabling a precise localization and classification of relevant surgical elements in the scene.

The segmentation masks are then analysed to find tool-tissue pairs among the detected tools and TTI regions to determine whether physical contact is occurring (i.e., *isolate* the TTI). It is worth noticing that the isolation process is needed especially in case multiple tool and tissue portion masks are detected and identified by YOLO.

The isolation process, whose architecture is depicted in Fig. 2, is based on a fine-tuned Vision Transformer (ViT) classifier [21], which analyses all possible tool-tissue pairs in the segmented frame to discriminate if such pairs actually imply a physical contact or not (i.e., if they are TTIs or not). To this end, a region of interest (ROI) enclosing the candidate tool-tissue pair is first computed as the smallest rectangle fully containing both the tool and tissue masks. This ROI is then provided to a ViT-based classifier, together with the corresponding tool and tissue masks and the depth map of the same region, obtained using the pre-trained Depth Anything v2 network [22].

This two-stage approach, instance segmentation followed by contact classification, ensures that the system not only localises and identifies surgical tools and interactions but it is also able to isolate tool-tissue pairs actually in contact.

III. DATASET

The dataset videos used for fine-tuning the YOLOv11-Seg network and the ViT-based classifier have been originally collected in [14] and are composed by around 2600 video clips of LC procedure, each 5 seconds long, captured at 30 frames per second. These video clips come from 289 publicly available LC videos of 153 surgeons from 136 institutions spread among 57 countries worldwide. The dataset videos have been manually annotated for the proposed work by surgeons at the University of Toronto. The annotations include tool masks and types, masks of the interaction areas (i.e., tissue masks) and type of ongoing interactions. Frames have been annotated when interaction starts, when it ends and in a varying number of frames in between. The total number of annotated frames in the dataset is 13.965. The annotated tool types are 8: Dissector, Scissors, Suction, Grasper, Harmonic, Bipolar, Cautery (hook, spatula), Stapler. Interaction types are 7: Coagulation, Retract and grab, Blunt dissection, Energy - sharp dissection, Cut - sharp dissection, Staple, Other. The number of instances in the dataset of the various tool and tissue types are reported in Table I. The dataset composition reflects the imbalance of tools/interaction types specific to the target surgical procedure. All these annotations (masks, tool types and interaction types) come as a set of JSON files, one for each video clip. The dataset has been split into

Tool types		Interaction types	
Type	Number	Type	Number
Dissector	2688	Coagulation	178
Scissor	148	Retract and grab	6331
Suction	344	Blunt dissection	3618
Grasper	6364	Energy - sharp dissection	2116
Harmonic	76	Cut - sharp dissection	530
Bipolar	273	Staple	72
Cautery (hook, spatula)	3731	Other	758
Stapler	93		

TABLE I: Number of instances of tool types and interaction types in the dataset.

training and test videos, respectively 75% and 25% of the total videos. Splitting on videos rather than on frames ensures to avoid contamination biases in the models.

IV. EXPERIMENTAL SETUP AND RESULTS

To assess the performance of the proposed approach, YOLOv11m-Seg (22M parameters) has been trained with the following hyperparameters: AdamW optimizer with initial learning rate equal to 1e-3 and final learning rate 1e-5, batch size equal to 16 frames, and 50 training epochs with early-stopping. To avoid model overfitting the patience was set to 10 epochs. The multi scale parameter was set to true, so to render the model more accurate with multiple sized image during inference. Detection-specific settings included Intersection over Union (IoU) threshold equal to 0.8, confidence threshold equal to 0.25, standard data augmentation (flip left-right, scale, translation, hue/saturation/brightness adjustment, mosaic). We choose to use this hyperparameter configuration as it show the best results on the test dataset. For ViT-based classifier we used ViT B/16 pre-trained model with Adam optimizer, learning rate 1e-4, L2 regularization weight 1e-4, 40 training epochs with early-stopping with 10 epochs patience. The inference time for each frame is about 0.1 seconds on a Nvidia Tesla P100 GPU (i.e., 10 fps prediction throughput).

A set of metrics have been collected to assess the performance of each step of the TTI understanding process on the test dataset. In the TTI detection step, performed by YOLOv11-Seg, an mAP50=0.9141 and an mAP50-95=0.8395 for segmentation of tool and tissue masks. The normalized confusion matrix of the TTI identification step, is reported in Fig. 3. The average accuracy is 82.46%.

The performance of the full end-to-end proposed pipeline has been assessed against the approaches described below. First, we have considered an intersection-based classifier, which decides a candidate tool-tissue pair being a TTI if there is any intersection among the respective masks. Ideally, this approach is suitable for the TTI isolation task, because the dataset has been annotated with masks of tools and tissues actually in contact. Therefore, assuming perfect learning, YOLO should detect and identify tissue and tool masks if and only if there is an involved TTI. This is clearly

not the case and false positives can occur in the isolation process. To mitigate false positives, a depth-based variant of this classifier has been also implemented. This variant considers a candidate tool-tissue pair being a TTI if there is an intersection and the difference between the median depth of the masks in the intersection area is less than a normalized threshold d_{th} . The criticality of this method is represented by its strong dependence on the choice of the appropriate threshold d_{th} . To minimize the number of false positives and improve precision, $d_{th} = 0.1$ has been chosen empirically.

For further comparison, we have investigated the impact of depth on the classifier described in Sect. II, and illustrated in Fig. 2, by eliminating the input channel conveying depth information and repeated the training. This benchmark can be seen as an ablation study to assess the value added by depth information.

For all the approaches under consideration, we evaluated the end-to-end pipeline in terms of confusion matrix, accuracy, precision, recall, and F1-score. Specifically, the entries of the confusion matrices in Tab. II are computed as follows.

- Correct TTI predictions are those test pairs whose masks are correctly detected and identified and, subsequently, correctly isolated.
- Incorrect TTI predictions are those pairs:
 - whose masks are correctly detected and identified (i.e., the masks are also present in the ground-truth) but incorrectly classified as TTIs in the isolation step;
 - whose masks are incorrectly detected and/or identified (i.e., the masks are not present in the ground-truth and/or the identification is wrong) and are incorrectly classified as TTIs in the isolation step.
- Correct No-TTI predictions are test pairs:
 - whose masks are correctly detected and identified (i.e., the masks are present in the ground-truth) and are subsequently correctly classified as No-TTI;
 - whose masks are incorrectly detected and/or identified but correctly classified as No-TTI in the isolation step.
- Incorrect No-TTI predictions are associated to tool and tissue masks correctly detected and identified but incorrectly classified as No-TTI in the isolation step.

The results from the confusion matrices show that the proposed TTI isolation approach is able to get both the highest number of correct predictions (i.e., the sum of the elements on the main diagonal of the matrices) and the lowest number of incorrect predictions (i.e., the sum of the elements on the secondary diagonal of the matrices) compared to all the other benchmarks.

It is worth noticing that, having included in the computation of Incorrect TTI and Correct No-TTI predictions also the pairs whose masks are incorrectly detected and/or identified, i.e., elements that are not present in the ground truth but are incorrectly introduced by YOLO (a varying number across the considered validation approaches), the sum of all the elements of the confusion matrices may change depending

TTI Isolation approach	Confusion matrix		
Intersection	True	Predicted	
		TTI	No-TTI
		312	33
Intersection + depth	True	Predicted	
		TTI	No-TTI
		1	344
ViT-based	True	Predicted	
		TTI	No-TTI
		336	9
ViT-based + depth (ours)	True	Predicted	
		TTI	No-TTI
		331	14
		76	305

TABLE II: End-to-end confusion matrices of the proposed pipeline with different TTI Isolation approaches.

on the considered approach. This choice is necessary to correctly evaluate the end-to-end performance of the method and does not change the overall assessment of the end-to-end performance of the approaches.

The metrics in Tab. III show that the isolation process based on ViT with depth information outperforms all the other benchmark approaches in terms of accuracy and F1 score. Although the intersection-based approach gains similar accuracy, it underperforms in terms of recall, which is particularly important for surgical applications, as we want to capture as much true TTIs as possible. Moreover, while intersection-based approach does not benefit of depth information (the distance threshold parameter is difficult to tune), it is possible to notice that depth improves consistently the performances of the ViT-based approach in terms of accuracy, precision and F1 score, while the recall is not significantly smaller.

It is worth commenting that, although the average performance of the intersection method appears quite satisfactory, we observed that it does not perform well on frames showing complex scenes, with instruments acting closely, because masks are spatially proximate or overlap. The generalizability of this method has been therefore considered limited, especially in view of an extension of the method to triplet-based modeling of tool-tissue interaction. Note also that, the high precision of the intersection method with depth is due to fact that it is overconservative in clasifying a TTI. As a matter of fact, the confusion matrix shows that only one TTI was predicted and it was a true TTI, thus the 100% precision.

Examples of TTI understanding are illustrated in Fig. 4-5. In Fig. 4 a single TTI is present, with a single tool mask and a single tissue mask detected: the pipeline correctly isolates the single TTI and identifies both the instruments and the type of TTI. In Fig. 5, instead, two TTIs are present: both have been correctly detected, together with the relative instruments and tissue masks, identified and isolated.

V. CONCLUSIONS

In this study, we introduced Tool-Tissue Interaction (TTI) understanding, i.e., an AI task for detection, identification

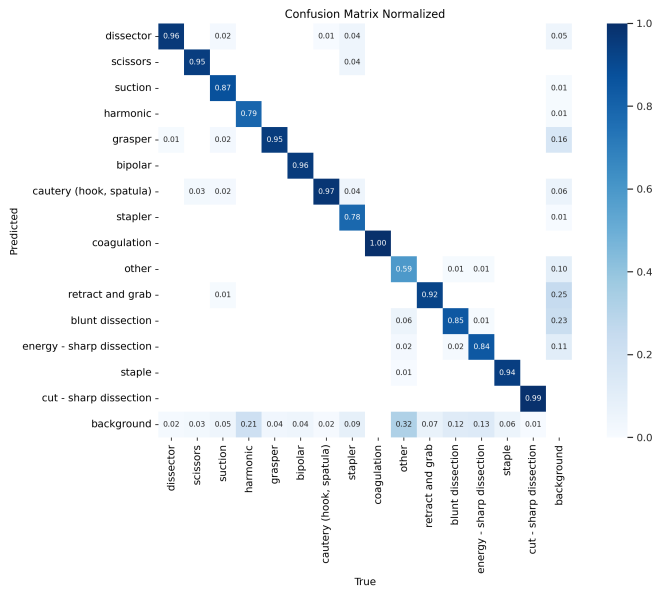
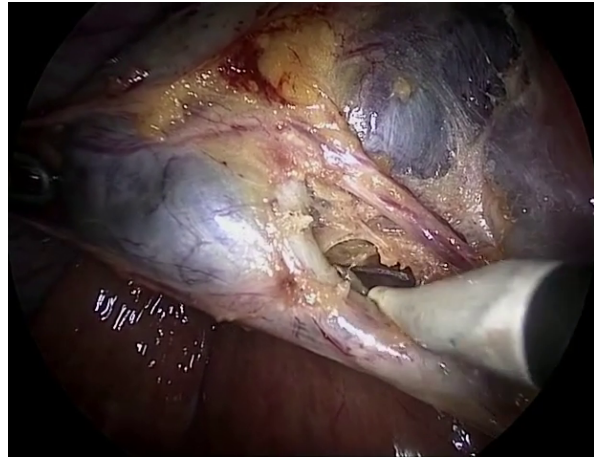
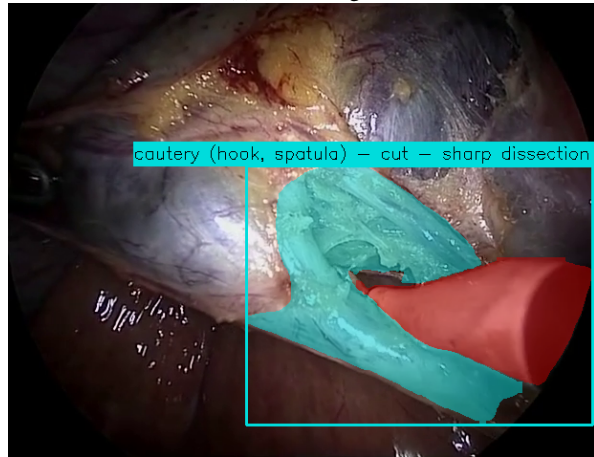


Fig. 3: Normalized confusion matrix from YOLOv11 in TTI Identification.



(a) Raw image.

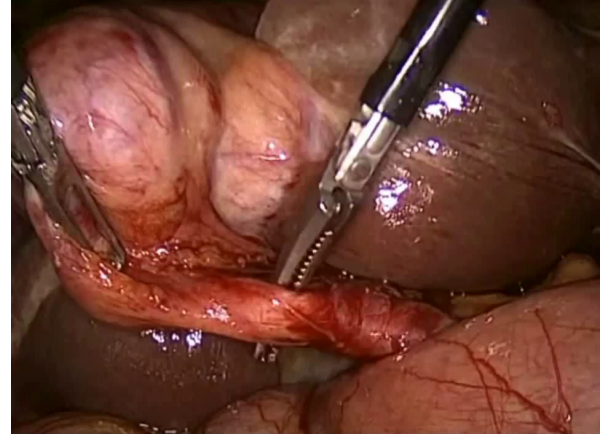


(b) Pipeline output.

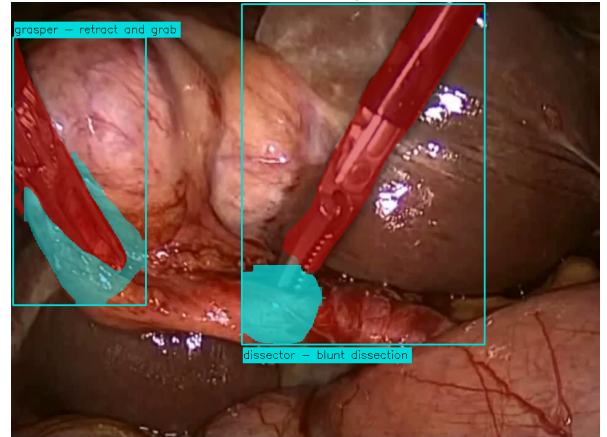
Fig. 4: Example of pipeline output in case of one TTI. Red masks are detected tools, light-blue mask represents the detected TTI region where contact occurs. The box represents the tool-tissue pair isolation corresponding to the TTI identified by the box label.

TTI Isolation approach	Accuracy	Precision	Recall	F1 Score
Intersection	87.33%	84.09%	90.43%	87.14%
Intersection + depth	52.62%	100%	0.29%	0.58%
ViT-based	81.96%	72.26%	97.39%	82.96%
ViT-based + depth (ours)	87.61%	81.33%	95.94%	88.03%

TABLE III: End-to-end results of the proposed pipeline vs TTI Isolation approach.



(a) Raw image.



(b) Pipeline output.

Fig. 5: Example of pipeline output in case of two TTIs in the same frame.

and isolation of instrument contact instances in surgical videos. This work aims at filling a gap in existing approaches linking surgical scene understanding and action detection by explicitly modelling physical contacts between tools and tissues. The proposed contact-aware TTI framework is directly applicable to surgical robotic systems, where reliable detection of physical tool-tissue contact is essential for safe autonomy and shared control. By leveraging depth-aware reasoning to distinguish true contact from mere proximity, the method provides actionable perceptual cues that can inform contact-aware control, safety monitoring, and interaction-aware task execution. Its real-time performance supports

intraoperative deployment. The proposed pipeline is based on a YOLOv11-Seg model for detection and identification, and a ViT-based classifier, leveraging depth information generated by Depth Anything v2 network, for isolation. The dataset used for training and testing these networks has been taken by publicly available LC videos, manually annotated with tools and contact areas masks and interaction types.

The proposed approach has been compared against learning-based and deterministic TTI isolation approaches, obtaining better end-to-end results, particularly in terms of accuracy and F1 score together with an higher number of good predictions and a lower number of wrong predictions with respect to all the other tested approaches.

Future work will focus on introducing temporal-based TTI understanding as well as introducing depth as additional information also for the detection and identification process. Future research will aim at devising weakly-supervised version of the proposed approach to reduce the manual annotation effort is also planned.

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